

MATH1014 Algebra

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Contents

Preface	v
0.1 What is Algebra?	v
0.2 Table of Symbols	v
1 Arithmetic	1
1.1 Modular Arithmetic	1
1.2 Some Number Theory	1
1.2.1 The Euclidian Algorithm	1
2 Fields and Rings	3
3 Equivalence Relations	5
4 Congruence of Integers	7
5 Vector Spaces	9
5.1 Subspaces	9
5.2 Span	9
5.3 Direct Sums	9
6 Linear Maps	11

Preface

THESE NOTES follow, and are meant to cement the course *Algebra*, taught at the University of Western Australia by Dr. Gordon Royle in Semester 2 of 2026. Where needed, the content of lectures is supplemented by *Linear Algebra Done Right* by Sheldon Axler.

0.1 What is Algebra?

I DO NOT YET KNOW ENOUGH TO SAY.

0.2 Table of Symbols

Symbol	Meaning
$\forall$	For every <i>or</i> for all
$\exists$	There exists
$\in$	In <i>or</i> is an element of
$\subset$	Is a proper subset of
$\subseteq$	Is a subset of
$\cup$	Union with
$\cap$	Intersection of
$\setminus$	Not <i>or</i> without
$\emptyset$	Empty set
$\sim$	Relates to
$>$	Greater than
$<$	Less than
$\geq$	Greater than or equal to
$\leq$	Less than or equal to
$ n $	The absolute value of n
$\square$	It has been proven

Chapter 1

Arithmetic

1.1 Modular Arithmetic

1.2 Some Number Theory

1.2.1 The Euclidian Algorithm

Chapter 2

Fields and Rings

Chapter 3

Equivalence Relations

Chapter 4

Congruence of Integers

Chapter 5

Vector Spaces

5.1 Subspaces

5.2 Span

5.3 Direct Sums

Chapter 6

Linear Maps